



Private browsing not so private?

A Stanford University research points out that the private browsing feature in many popular browsers retain information when they should not



World of Warcraft running on Windows in VirtualBox running on Linux

run even the most recent games such as Portal 2, Dragon Age II, Crysis 2 and RIFT in addition to popular titles such as World of Warcraft, StarCraft II, and Half-Life 2.

Furthermore, there's a greater push towards releasing games on Linux now. The Indie gaming community ran the Humble Indie Bundle promotions, which were collections of cross-platform games at pay-what-you-want pricings, and Linux users were the largest contributors to sales.

Even better news lies ahead, as new game stores for Linux users are being launched. Gameolith, a Linux-only game store just launched with five titles, while Desura, an Indie-gaming-centric digital distribution store similar to Steam, is working towards a Linux offering. Speaking of Steam, rumours are abuzz that Valve is working on Steam for Linux, after they've already released a version for Mac OSX.

Currently, games mostly use the

DirectX framework which is Windows-only, but OpenGL is once again starting to be a favourable technology. The apps themselves are changing in such a way that the OS platform might not matter as much as it used to.

...the apps themselves...

While Windows has stuck mostly to PCs, the scope of computing has gone far beyond that, and today an application developer needs to target more than just Windows if they want to reach the broadest audience. This means more apps developed using cross-platform technologies and toolkits, and thus increasing portability to Linux as well.

While DirectX only runs on Microsoft's Windows and XBOX platforms, OpenGL, its open alternative, is supported on Windows, Linux, Mac OSX and PlayStation 3. Furthermore OpenGL ES, a derivative of OpenGL, is supported on Android, iOS, Symbian, Bada, WebOS and BlackberryOS. WebGL, the new standard that enables 3D content on the web, is also based on OpenGL ES. As more game developers target mobile and web gaming, OpenGL will once again become the platform of choice.

A second change that is coming to applications is how they're delivered. A large number of great applications now run directly in the browser, making the underlying OS meaningless as long as it runs a browser that supports the latest web standards.

Another new mechanism that makes the OS running on a device inconsequential is streaming applications. A brilliant example of this would be OnLive, a new gaming system

that streams games to your computer instead of running the game on your computer. Essentially, the game is played on its servers and a video of the game is streamed to your computer. Any mouse movements, key presses or gamepad data is sent to the game over the net. This way one can run a game in full HD without the need for a powerful graphics card. In fact, you can also use a netbook! With improving internet speed, such technologies will only become more powerful. Such mechanisms are also available for using applications rather than games.

Yet another change is coming to applications, and that is virtualisation. Running Windows in a Linux environment means that any Windows application will run perfectly. There's some evidence that game developer Ubisoft is playing around with this idea, as they've posted a video that demonstrated how a single computer can run two games on two monitors at the same time using two graphics cards. The OS they're using is Linux while the games are running on a Windows running in a virtual machine (VM).

With newer means and mechanisms for application and game delivery, one major barrier, that is lack of major applications / games on Linux will also pass.

...content in vain?

The future of any OS though, depends on how far into the future we look. The current architecture of operating systems was developed for computers that had a single processor, and has over time expanded to include better support for multicore computing.

MIT researchers have shown that Linux is capable of scaling to as many as 48 cores with small modifications, so we aren't yet ready to abandon all OSs, there's still time.

That however is where the future lies. As the number of cores increase into triple digits and beyond, a new architecture of operating systems will be needed. Such operating systems will, in all likelihood, more closely resemble the distributed computing systems of today than either Windows or Linux. ■



Angry Birds running in the browser on Windows, but it could as easily be Linux or Mac OSX